

Hello, my name is Greg Dyer, Associate Professor, Toro College of Osteopathic Medicine. This video was designed to review the medications used by both patients in the upper and lower extremity core cases. Both patients have a disease mechanism that involves pain and inflammation. In both cases, they self-administer ibuprofen and NSA to relieve pain and inflammation. Both patients are being treated with a RAS inhibitor, lecinopril, to treat their symptoms of hypertension. The focus of this video will be to highlight the mechanisms of action used, toxicity, and possible drug interactions that may occur to enable you to best evaluate your patients. The patients described in the upper extremity core also self-medicates with acetaminophen and perhaps alcohol, so the possible drug interactions will be briefly discussed. Pharmacotherapy of an inflamed joint includes palliative therapy to reduce pain and inflammation. Drugs commonly used are NSAIDs and acetaminophen. This is why they were used in these patients. These drugs will be the focus of this review. Not covered in detail in this review is the use of intra-articular injections of glucocorticoids into the inflamed joint. This is a common procedure to both temporarily reduce inflammation, as glucocorticoids are the most efficacious anti-inflammatory drug available and can be diagnostic to help ascertain the disease mechanism. For example, if after the injection, pain is relieved and resolves, then the underlying inflammatory mechanism is the focus of the diagnosis. However, if after the injection, there is only temporary pain relief, there is a suggestion of a structural damage that has to be addressed. In both cases, the option was not utilized, so we won't focus on this in this review. What we will focus on is NSAIDs. NSAIDs have an important role at reduction in inflammation and pain. They also are important at any fever medications as well as they have effects on platelets. The mechanism of action of NSAIDs and aspirin and to some extent acetaminophen involves starting with the damaged tissue and the activation of phospholipase A2. Phospholipase A2 is an enzyme that produces arachidonic acid. Arachidonic acid is the common precursor that feeds into enzymes that produce pro-inflammatory products. It's another issue, too, is the diet affects the activity of the style of arachidonic acid that exists in your body. There's two styles that are produced from your diet, omega-6 style and omega-3 style. This is the idea of using diet to regulate the inflammatory cascade. Again, this won't be the focus of this discussion, but this is the precursor that is affected by diet. Now, arachidonic acid is converted by two primary enzymes, lipoxygenase and cyclooxygenase. Both enzymes produce pro-inflammatory products. The focus of this will be cyclooxygenase. Cyclooxygenase is the enzyme that's inhibited by NSAIDs, aspirin, and acetaminophen. That's shown on this slide. So, arachidonic acid is converted by cyclooxygenase into a common precursor, PGH2. PGH2, then, is converted in various cells in different locations in the body to different styles of prostaglandins and thromboxanes. And they're shown at the bottom here. So, when we look at the different styles of prostaglandins and thromboxanes produced, it depends on the location and the enzyme. There is a difference between COX-1 and COX-2, not in what they produce and how they're activated. For example, COX-1 is inducible by cytokines and growth factors. It has a role in inflammation, fever, and pain. It's the target, then, for most anti-inflammatory drugs. It's located in both inflammatory and endothelial cells, as well as other cells, like colon polyps. High-yield issues with COX-2 inhibition. Now, ibuprofen is a non-selective. It inhibits both COX-1 and COX-2. So, you're going to have both activity of COX-1 and COX-2. There are selective NSAIDs, Celebrex, Celecoxib, etc., which just primarily work on the COX-2 enzyme. The high-yield issues about COX-2, COX-2 enzyme is important for mitogenesis and growth, female reproductive regulation, bone formation, renal function, and effects on platelets. COX-2 is a constitutively expressed homeostatic enzyme. It's expressed in all tissue, found in platelets, and it's also found in the gut, where it provides a protective role for your gut lining. Or, when we think of COX-2, enzyme products, we think of that as GI mucosal protectant, increased platelet aggregation through phenolmoxane A2 synthesis, and then, again, renal homeostasis. So, when you look at this, the products of cyclooxygenase are both important for regulating inflammation and creating the homeostatic milieu that allows us to survive. In the

inflammatory side, the arachidonic acid products, thromboxane, prostaglandins, leukotriene, all play a role in stimulating inflammation and producing the ability of that joint to heal and temporarily lock it down so that it doesn't cause further damage. The problem is, is the inflammatory cascade, it can go unchecked and can create more damage. So, more often than not, it's important to reduce the inflammatory cascade so that it doesn't create further damage to the joint structure. These products, thromboxane, prostaglandins, leukotrienes, produce immune cell attraction, activation, and modulation.

They increase capillary fluid flow, so there's increased vascular permeability.

This allows for access to the joint to the immune players, as well as immobilizing the joint so that no further damage occurs. Increases endothelial cell gaps and swelling and secretions. Also causes phagocyte attraction, activation, T cell modulation. This is the inflammatory cascade that is addressed when you use NSAID. So, using an NSAID reduces the production of these excitatory inflammatory mediators so you get less inflammation. In addition, arachidonic products, these prostaglandins, shown on this slide, can also stimulate the activity of that primary nociceptive neuron. And you can look here on this side, PGE2 activates its receptor, which then stimulates activity inside the cell, which makes it easier for this nociceptive primary afferent to fire. It decreases the threshold for activation of the nociceptive neuron, so that that neuron then can send a signal up to your brain where it's perceived as pain. Inhibiting the production of this prostaglandin, therefore, can reduce the ability of that neuron to fire so readily.

Things to monitor in your patient depend on what the endogenous prostaglandins do. In the CNS, if there's ringing in your ears or dizziness, obviously you're going to stop the use of ibuprofen and use an alternative. Both patients had no serious cardiovascular issues, , but they did have hypertension.

heart failure, edema, etc., but they did have hypertension. And the combinatorial effects of NSAIDs and Lisinopril will be then a focus of the discussion moving forward. Both patients were on ibuprofen. They were relatively young. GI issues are definitely an important feature, especially in the elderly. So, you're going to have to monitor their symptoms for GI. Are they anemic?

Do they have pain, etc. Hematologic thrombocytopenia, neutropenia, aplastic anemia, decrease platelet aggregation occur. So, in these both patients, you're going to make sure you monitor their blood values using CBCs.

Liver function tests are commonly performed and should be evaluated in the context of NSAID use.

Pulmonary asthma, asthma doesn't occur often, but it can occur in some patients where you get NSAID-induced asthma is because the overproduction of leukotrienes as the arachidonic acid is shunted into the leukotriene pathway.

Rashes occur, may occur.

And then renal is another big target. So, two things to really focus on in these patients is to make sure because they have hypertension that they're being effectively controlled and there's no combinatorial issues, especially on renal function. And then renal is also a big deal. As these patients are taking ibuprofen and they're taking Lisinopril, this can have a dual effect on renal function we'll discuss.

Prostaglandins play several roles in renal function. And one of the easiest to explain roles is the role in maintaining GFR.

Think of the times of volume contraction, you become hypovolemic or you have heart failure or your pumping action is not sufficient to keep the filtration pressure up in the renal vascular beds.

The mechanisms that prostaglandins provide is to cause vasodilation of this incoming blood vessel.

This vasodilation then keeps up glomerular filtration pressure at times of hypovolemia.

This is a survival mechanism. So, think of prostaglandin release as being promoting a balance of N equals out in the nephron.

And when you inhibit that, you inhibit those vasodilatory prostaglandins, less flow goes in and it can affect GFR. And usually this is not a big deal in healthy individuals as there's enough volume to keep the pressure up in the nephron.

It only becomes an issue if they have heart failure or hypovolemic for some reason and it could be even sweating and becoming dehydrated.

These drugs do cause acute renal failure in the smallest percentage of people and can cause interstitial nephritis. So, monitor your renal functions in both patients in this case. Acetaminophen is a mild to moderate pain.

It's not associated with inflammation. It's thought to work more centrally than peripherally.

And so, it's associated with less of the platelet inhibition effects as well as less anti-inflammatory effects as compared to NSAIDs or aspirin. The advantage of it is it doesn't have, if your patient has any kind of bleeding disorders or has a potential for surgery or something like that, that acetaminophen can be used and not cause as much loss of the coagulation cascades. Now, one of the issues with the upper extremity patient was that she was kind of a moderately high level of alcohol consumption.

This is normally not a level associated with overtotoxicity.

However, any drinks more than a half a drink a day is associated with several poor outcomes ranging from cancer to hypertension to liver issues.

And the combination of acetaminophen and alcohol can be an issue.

Acetaminophen is metabolized pretty rapidly by conjugation reactions.

So, these conjugation reactions are phase two reactions where they add moiety on the acetaminophen which allows it to be charged and then excreted.

These conjugation reactions produce inner products.

About five percent of an acetaminophen is modified by Cytochrome P450 2E1.

And when 2E1 is modified, it produces a toxic free radical.

So, this free radical can bind to mitochondria and inhibit liver function and cause liver damage.

This is rapidly scavenger by scavenger pathways.

Again, it's modified by a conjugation reaction so that it's no longer a free

radical.

In the case of alcohol, alcohol induces 2E1.

So, if you use alcohol and you take acetaminophen, you're increasing the amount that gets converted into this free radical. And sometimes it can overcome the scavenging pathway. So, that's something to pay attention to this patient.

The hypertensive medication that both patients are using is lisinopril. And we look at the cascade here. It's down here. These are very common drugs used, very successful, very good drug, very limited toxicity profile. They can be used as a choice along with diuretics or calcium channel blockers, et cetera. The mechanism of action of lisinopril is it's an angiotensin-converting enzyme inhibitor.

So, it inhibits this enzyme right here. Angiotensin 1 isn't converted into the act of angiotensin 2. And so, that you don't get the product that causes vasoconstriction, angiotensin 2, aldosterone secretion which means sodium retention, vasoconstriction means increased peripheral resistance and increased blood pressure. By stopping it upstream, at this level, you don't get the preceding cascade.

Now, angiotensin 2-converting enzymes also break down bradykinin into its inactive product. Bradykinin is also associated with an increase in prostaglandin synthesis which can cause vasodilation.

And it does contribute a bit to lowering blood pressure. So, the concomitant use of NSAIDs along with the angiotensin-converting enzyme inhibitors can reduce its ability.

Not in everybody, but it can occur. And it's not a big effect. I've seen papers 16% reduction in ability for ACE inhibitors to reduce blood pressure, something like that. As inhibitors are associated with the toxicities that weren't exhibited in these patients, severe hypotension, dry cough, angedema caused by the bradykinin and the substance P. Anybody on an ACE inhibitor would be monitored for hyperkalemia.

Aldosterone production is inhibited.

So, less aldosterone means less potassium excretion and less sodium retention. Contraindication, acute renal failure.

Anytime you reduce blood flow going to that nephron, it could drop GFR past a certain level which becomes detrimental to your patient. It is a tritigen, so avoidant second and third trimester of pregnancy.

The drug interactions, potassium sparing diuretics can cause hyperkalemia, and NSAIDs impairs the effects of ACE inhibitors. The thing that you should monitor in this patient is the patient's GFR. Here's what normally happens in renal function. So, we've already talked about the glomerulus, that you have vasodilatory factors, prostaglandins, increasing the flow in, and then you have constrictive flows that reduce the flow out. These are survival hormones that are released during times of hypovolemia to keep GFR up. This is how you maintain normal GFR when your blood pressure can fluctuate, when your blood volume can fluctuate.

NSAIDs reduce the inflow, so they can drop GFR by themselves.

Ras blockers, lisinopril, preventing angiotensin 2, prevent the invasive constriction. So, here you can see that the outflow is increased. And in normal people, this is not an issue. Normal, healthy individuals who are normal volume,

who have normal renal function, this tends not to be an issue. It's not something that you have to worry about. So, in both these patients, nothing indicated that there was an issue, although you do have to measure renal function. NSAIDs and Ras blockers increase wrist to drop GFR. So, it's something you pay attention, especially if they have any other entry issue going into the nephron. They're on a diuretic.

They have heart failure. They become hypovolemic. Then this issue can be more pronounced, and you can drop GFR to an extent where it causes renal damage.

Diuretics decrease water volume. So, they're another way to treat hypertension. These two papers I show here, discuss in detail, is that avoid using all three together.

When you use all three together, you drop the influx, so there's less filling pressure of the nephron, so less pressure.

And then you use an NSAID, which prevents this dilation from occurring. And you use an ACE inhibitor or an ARB, which prevents this from constricting.

All these three things together, one, two, three, drop GFR. And in enough people, it's suggested to avoid this combination. This is this sort of triple whammy of decrease in GFR. So, avoid this in all patients.

Generally speaking, NSAIDs and Ras blockers used together are okay in people who don't have heart failure or any kind of volume reduction. However, in most people, you want to avoid diuretics, NSAIDs, and angiotensin-converting enzyme inhibitors, or ARBs.

Avoid those all together.

The last thing that was covered in patient one, the upper extremity, she was on human, regular human insulin.

There's different types of insulin. She has type 1 diabetes, which is a disease associated with the destruction of their pancreatic islet cells that produce insulin. So, the patient has to add back insulin, timed with meals.

And regular insulin simply is a form of insulin that's released in a sustained way and peaks about three hours after.

So, it's different formulations.

There's rapid acting insulin. There's long acting insulin. And these are basically polymerized so that they release the insulin at different rates. And the combinations are commonly used. Usually, the insulin is timed with the meals so that the postprandial glucose can be brought back down. All right. So, that's what we have. So, we've covered the 10-minute farm review, NSAIDs, acetaminophen, Ras inhibitors, and regular insulin. All right. Good luck.